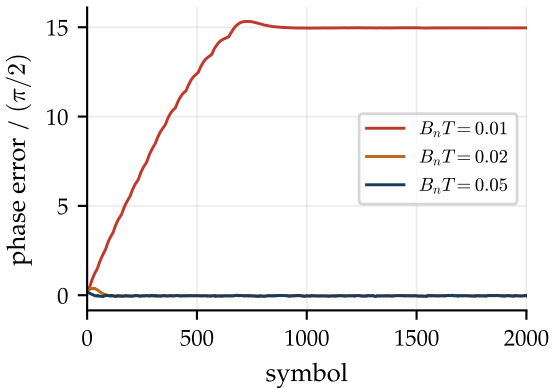


(a) pull-in of a QPSK DD loop, $\nu = 0.008$



(b) NCO frequency (25-symbol average)

